SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics.* (BTL 5)

Assessment Tool Weightage:20%

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution

Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

3. Motion Analysis Decision Problem

Problem Statement

A traffic monitoring system needs to analyze the movement of vehicles from video footage. The system has two important requirements:

1. **Accurate motion direction detection** – It should correctly identify whether vehicles are moving left, right, forward, backward, etc.
2. **Robustness in dense traffic** – It should continue working reliably when many vehicles are present close to each other.

Two techniques are being considered:

- **Optical Flow**
- **Motion Estimation**

The objective is to compare both methods and select the most suitable method for **high-density traffic conditions**.

Optical Flow

Optical Flow is a computer vision technique that estimates how pixels or visual features move between two consecutive video frames.

For example, consider two frames:

Frame 1 Frame 2

🚗 🚗 🚗 🚗

🚗 → 🚗 →

🚗 🚗

By comparing the positions of pixels between the frames, Optical Flow generates **motion vectors**.

Advantages

- Provides detailed motion information.
- Can determine the direction of individual objects.
- Can estimate the speed and direction of movement.
- Useful when vehicles are clearly separated.
- Provides dense motion information across the image.

Disadvantages

In highly crowded traffic, many objects are close together.

For example:

🚗🚗🚗🚗🚗🚗

🚗🚗🚗🚗🚗🚗

🚗🚗🚗🚗🚗🚗

The movement of neighboring vehicles can overlap. This can produce:

- Noisy motion vectors
- Incorrect directions
- Ambiguous object boundaries
- Less reliable tracking

Therefore, although Optical Flow is very detailed, its performance can decrease in highly crowded scenes.

Motion Estimation

Motion Estimation determines the movement between consecutive frames, often by comparing blocks, regions, or important features.

Instead of analyzing every pixel independently, it can estimate the movement of larger regions.

For example:

Frame 1 Frame 2

[Vehicle Group] → [Vehicle Group]

The system estimates the overall movement of the region.

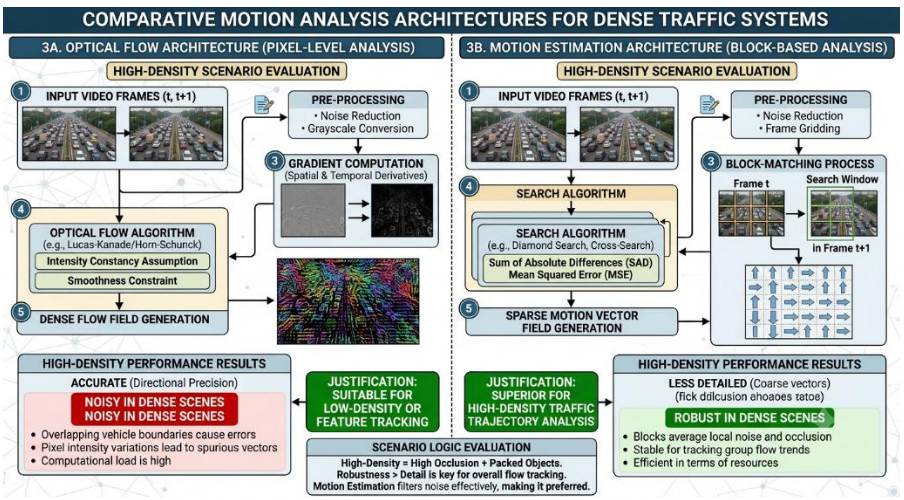
Advantages

- More stable in crowded scenes.
- Less affected by individual pixel-level noise.
- Computationally efficient in many implementations.
- Suitable for estimating overall traffic movement.
- More robust when vehicles overlap or partially block one another.

Disadvantages

- Less detailed than Optical Flow.
- May not capture very small individual movements.
- Individual vehicle motion may not be represented precisely.
- Direction information can be less granular.

ARCHITECTURE DIAGRAM:



Comparison of Optical Flow and Motion Estimation

Feature	Optical Flow	Motion Estimation
Motion detail	Very high	Moderate
Direction detection	Highly detailed	Stable
Dense traffic	Can become noisy	More robust
Individual movement	Excellent	Moderate
Noise sensitivity	Higher	Lower
Computational requirement	Can be higher	Generally lower
Suitable for isolated vehicles	Excellent	Good

Feature	Optical Flow	Motion Estimation
Suitable for crowded traffic	Moderate	**Excellent**

Scenario-Based Analysis

Consider a **busy city intersection during rush hour**.

Suppose there are 100 vehicles in the camera's field of view.

Some vehicles are:

- Very close to each other
- Moving in the same direction
- Overtaking each other
- Partially occluding other vehicles
- Changing lanes

If Optical Flow is used

Optical Flow will generate many motion vectors.

↗ → → ↘

→ 🚗 🚗 ←

↘ → 🚗 ↗

→ 🚗 ← →

Because vehicles are close together, the motion vectors may become noisy or conflicting.

This can cause incorrect interpretation of vehicle movement.

If Motion Estimation is used

The system can estimate the movement of larger regions:

Vehicle Group

↓

Overall movement

↓

→

Even if individual vehicles overlap, the overall traffic movement can remain relatively stable.

Therefore, in this particular scenario, **robustness is more important than extremely detailed motion information**.

6. Final Decision

✅ **Recommended Method: Motion Estimation**

Motion Estimation is more suitable for high-density traffic conditions.

Justification

The question specifically requires:

Accurate motion direction detection **and** robustness in dense traffic.

Optical Flow provides more detailed motion information, but it can become noisy when many vehicles are close together.

Motion Estimation provides slightly less detailed information, but it is **more stable and robust in crowded scenes**.

Therefore:

High-Density Traffic

↓

Many Vehicles

↓

Overlapping Motion

↓

Optical Flow → More Noise

↓

Motion Estimation → More Stable

↓

BEST

Important point

If the traffic were **low-density**, with vehicles clearly separated, Optical Flow could be the better choice because its detailed motion information would be valuable.

However, because this problem specifically emphasizes **high-density traffic**, **Motion Estimation is the better decision**.

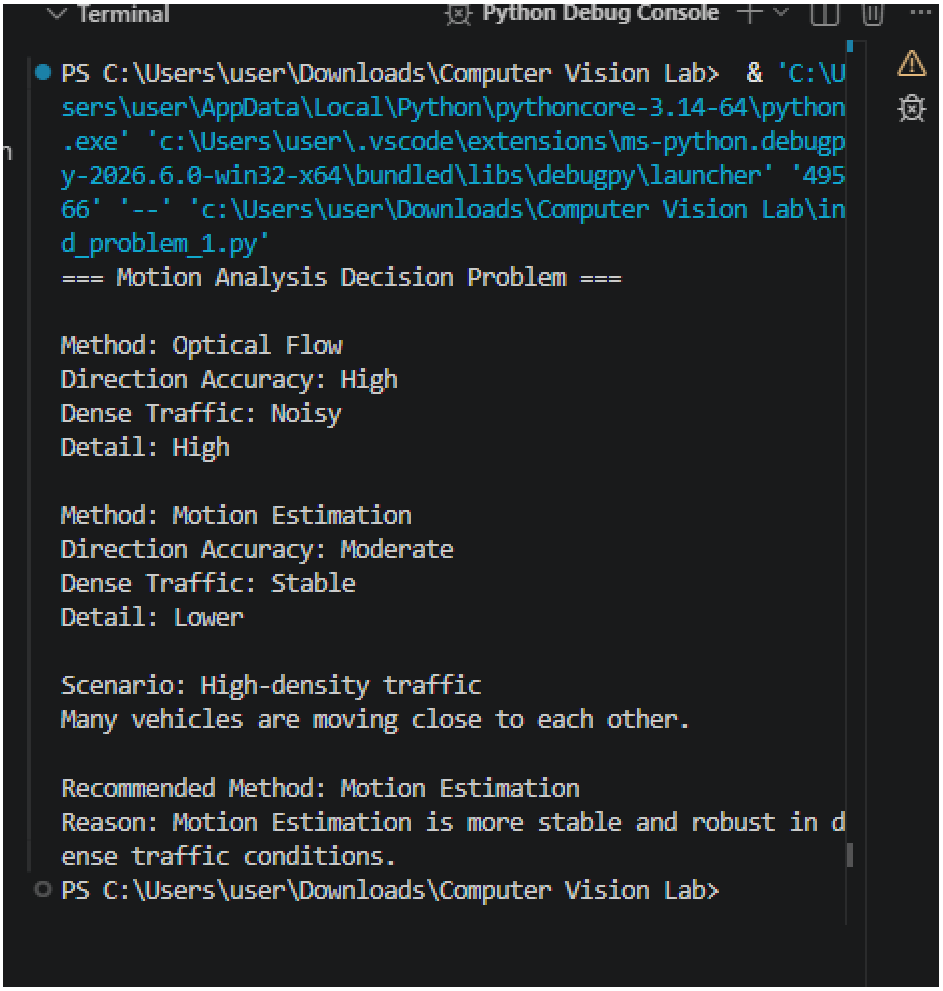
Python Program

```
# Motion Analysis Decision Problem


print("==============================================")

print("     MOTION ANALYSIS DECISION PROBLEM")

print("==============================================")


# Define the characteristics of both approaches

methods = {

    "Optical Flow": {

        "accuracy": "High",
```

```
        "dense_traffic": "Noisy",
        "detail": "High"
    },

    "Motion Estimation": {
        "accuracy": "Moderate",
        "dense_traffic": "Stable",
        "detail": "Lower"
    }
}

# Display comparison
for method, values in methods.items():

    print("\nMethod:", method)
    print("Direction Accuracy:", values["accuracy"])
    print("Dense Traffic Performance:",
          values["dense_traffic"])
    print("Motion Detail:", values["detail"])


# Traffic scenario
traffic_density = "High"
```

```
print("\nTraffic Density:", traffic_density)


# Decision logic
if traffic_density == "High":
    best_method = "Motion Estimation"
    reason = (
        "It provides stable and robust motion estimation "
        "in dense traffic."
    )
else:
    best_method = "Optical Flow"
    reason = (
        "It provides detailed motion information "
        "when vehicles are clearly separated."
    )


print("\n---------------------------------------------")
print("Recommended Method:", best_method)
print("Reason:", reason)
print("---------------------------------------------")
```

OUTPUT:



```
Terminal                              Python Debug Console
PS C:\Users\user\Downloads\Computer Vision Lab>  & 'C:\Users\user\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\user\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '49566' '--' 'c:\Users\user\Downloads\Computer Vision Lab\ind_problem_1.py'
=== Motion Analysis Decision Problem ===

Method: Optical Flow
Direction Accuracy: High
Dense Traffic: Noisy
Detail: High

Method: Motion Estimation
Direction Accuracy: Moderate
Dense Traffic: Stable
Detail: Lower

Scenario: High-density traffic
Many vehicles are moving close to each other.

Recommended Method: Motion Estimation
Reason: Motion Estimation is more stable and robust in dense traffic conditions.
PS C:\Users\user\Downloads\Computer Vision Lab>
```

19. Video Analytics Pipeline

1. Problem Statement

A video analytics system must process video frames in real time.

The system has two strict requirements:

Requirement 1 — Accuracy

The system must have:

Accuracy ≥ 90%

This means that an accuracy of:

- 90% → ✅ Acceptable
- 91% → ✅ Acceptable
- 95% → ✅ Acceptable
- 89% → ❌ Not acceptable

Requirement 2 — Processing Time

The system must process a frame in:

Less than 100 ms

Therefore:

- 95 ms → ✅ Acceptable
- 99 ms → ✅ Acceptable
- 100 ms → ❌ Not acceptable
- 120 ms → ❌ Not acceptable

2. Given Pipelines

There are three possible pipelines.

Pipeline	Accuracy	Processing Time
Pipeline A	88%	80 ms
Pipeline B	91%	120 ms
Pipeline C	90%	95 ms

We need to determine which pipeline satisfies **both conditions simultaneously**.

3. Evaluation of Pipeline A

Accuracy

88% < 90%

Therefore:

❌ Accuracy requirement is not satisfied.

Processing time

80 ms < 100 ms

Therefore:

✅ Processing-time requirement is satisfied.

Final result

Accuracy → ✕

Speed → ✅

Pipeline A → REJECT

Although Pipeline A is fast, its accuracy is insufficient.

4. Evaluation of Pipeline B

Accuracy

91% ≥ 90%

Therefore:

✅ Accuracy requirement is satisfied.

Processing time

120 ms > 100 ms

Therefore:

✕ Processing-time requirement is not satisfied.

Final result

Accuracy → ✅

Speed → ✕

Pipeline B → REJECT

Pipeline B is accurate enough, but it is too slow for the required real-time processing.

5. Evaluation of Pipeline C

Accuracy

90% ≥ 90%

Therefore:

✅ Accuracy requirement is satisfied.

Processing time

95 ms < 100 ms

Therefore:

✅ Processing-time requirement is satisfied.

Final result

Accuracy → ✅
Speed → ✅

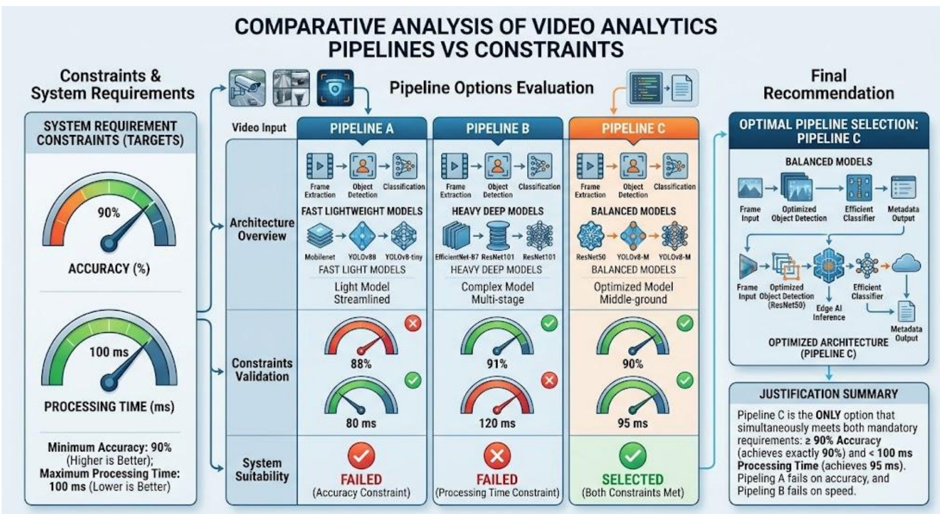
Pipeline C → ACCEPT

Pipeline C is the **only pipeline that satisfies both requirements**.

6. Comparison Table

Pipeline	Accuracy ≥ 90%	Time < 100 ms	Overall
A	✗ 88%	✓ 80 ms	✗ Reject
B	✓ 91%	✗ 120 ms	✗ Reject
C	✓ 90%	✓ 95 ms	✓ **Select**

ARCHITECTURE DIAGRAM:



Scenario-Based Explanation

Imagine the system is being used for **real-time traffic monitoring**.

The camera captures vehicles continuously.

The system must:

Camera

↓

Video Frame

↓

AI Processing

↓

Vehicle Detection
↓
Traffic Analysis
↓
Decision

For real-time operation, the system cannot be too slow.

Pipeline A

It processes the video quickly at **80 ms**, but its accuracy is only **88%**.

This could cause incorrect vehicle detections.

For example:

Actual vehicles = 100
Correct detections ≈ 88

This is below the required accuracy.

Pipeline B

It has excellent accuracy of **91%**, but takes **120 ms**.

For real-time traffic monitoring, this creates additional processing delay.

Required: < 100 ms
Pipeline B: 120 ms
↑
Too slow

Pipeline C

It provides:

Accuracy = 90%

Time = 95 ms

Both conditions are satisfied.

Therefore, it provides the required balance between **accuracy and speed**.

Final Decision

✅ **Pipeline C is the best pipeline.**

The decision can be represented as:

```
               Pipeline Options
                     |
        +--------------+--------------+
        |            |            |
     Pipeline A    Pipeline B    Pipeline C
     88%, 80ms      91%, 120ms      90%, 95ms
        |            |            |
   Accuracy FAIL    Time FAIL      Both PASS
        |            |            |
      REJECT        REJECT        SELECT
```

Justification

Pipeline C is the only option that satisfies **both constraints simultaneously**:

Accuracy:

90% ≥ 90% ✓

Processing:

95 ms < 100 ms ✓

Therefore, **Pipeline C is the optimal choice for the given requirements**.

Python Program

```
# Video Analytics Pipeline Decision Problem

print("==============================================")
print("       VIDEO ANALYTICS PIPELINE")
print("==============================================")

# Required constraints
MIN_ACCURACY = 90
MAX_TIME = 100

# Pipeline information
pipelines = {
    "Pipeline A": {
        "accuracy": 88,
        "time": 80
    },

    "Pipeline B": {
        "accuracy": 91,
        "time": 120
    },
```

```
    "Pipeline C": {
        "accuracy": 90,
        "time": 95
    }
}

# Evaluate every pipeline
for name, data in pipelines.items():

    accuracy = data["accuracy"]
    processing_time = data["time"]

    accuracy_ok = accuracy >= MIN_ACCURACY
    time_ok = processing_time < MAX_TIME

    suitable = accuracy_ok and time_ok

    print("\n", name)
    print("Accuracy:", accuracy, "%")
    print("Processing Time:", processing_time, "ms")

    if accuracy_ok:
        print("Accuracy Requirement: PASS")
    else:
        print("Accuracy Requirement: FAIL")

    if time_ok:
        print("Processing Time Requirement: PASS")
    else:
        print("Processing Time Requirement: FAIL")
```

```
    if suitable:
        print("Overall Result: ACCEPT")
    else:
        print("Overall Result: REJECT")



# Final decision
print("\n==================================================")
print("            FINAL DECISION")
print("==================================================")

print("Best Pipeline: Pipeline C")
print("Accuracy: 90%")
print("Processing Time: 95 ms")
print("Reason: Pipeline C satisfies both constraints.")
```

OUTPUT: